

# Coinnect Application Brief

UMasenso Hub Cohort 5 Pre-Incubation Program  
SmartTech entry from UM Tagum

## Application overview

We are developing Coinnect, a self-service kiosk for exchanging Philippine peso bills and coins and accessing cash-to-wallet services. We seek pre-incubation support to validate a campus use case, complete physical integration, and test a sustainable operating model. Coinnect is a partial physical prototype. We propose a supervised pilot at UM Tagum, subject to campus approval and demonstrated readiness.

## Problem

A student may have a large bill when a small payment needs coins, or have cash when a payment requires wallet funds. Staff may face the same mismatch. Finding someone willing and able to make change or provide cash-to-wallet service can add an extra step to an ordinary purchase. Nearby merchants are potential beneficiaries if customers can obtain suitable denominations before paying.

These are the problems we intend to investigate on campus, rather than findings from a completed survey. We have not established how often they occur, the time or cost involved, or how satisfied users are with current options. A kiosk is useful only if those inconveniences are frequent enough and users accept the service fee.

## Initial problem validation

We propose interviews with students, staff, and nearby merchants, followed by observation of common cash and wallet payment needs with permission. We will ask about the last actual incident, the alternative used, time spent, total charges, and willingness to use an on-campus kiosk. We will compare this evidence with a local review of existing services before finalizing prices or deployment.

## Team

| Member                  | Role        |
|-------------------------|-------------|
| Vince Ronald Maravillas | Team leader |
| Stunley Opeña           | Member      |
| Raziel Renz Salamat     | Member      |

All members: BS in Computer Engineering  
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# Solution

Coinnect brings cash denomination conversion and planned wallet access into one touchscreen kiosk. The first pilot priority is Philippine peso conversion: bill-to-coin, coin-to-bill, and bill-to-bill transactions, limited by available inventory. Local cash conversion is designed to work without internet connectivity. Wallet services require connectivity and verified payment processing.

## Customer experience

The user selects a service and reviews the available amount, output denominations, and fee before inserting cash or paying. The kiosk counts accepted funds, checks inventory, and guides the transaction to an outcome. Receipts or transaction references provide a record. When a transaction cannot be completed after funds are accepted, the design records the amount still owed and provides a claim reference for operator follow-up.

## Wallet access and later services

The codebase includes PayMongo integration for QR Ph cash-out and wallet disbursement flows intended for GCash and Maya cash-in. These integrations are not evidence of a commercial partnership or approved live service. A pilot would activate wallet transactions only after integration testing and appropriate provider and operating arrangements. Foreign exchange is a later capability and is outside the initial campus pilot.

## Prototype status and technical feasibility

| Evidence available                                                                               | What remains to establish                                                                   |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| React touchscreen flows, Python backend, and controller firmware are present in the repository.  | Repeatable end-to-end operation on the assembled kiosk.                                     |
| Code includes inventory control, transaction records, payment verification, and customer claims. | Correct behavior under jams, partial payouts, lost connectivity, and hard power loss.       |
| The design uses camera-based bill checks, cash handling mechanisms, and tamper controls.         | Measured bill recognition performance, dispensing reliability, and physical security tests. |
| The team confirms a partial physical prototype.                                                  | Documented hardware results and a supervised user trial.                                    |

## Development priorities

We will first establish consistent cash intake and payout, validate inventory reconciliation, and test recovery from interruptions. The design has no UPS, so a hard power cut must be part of physical testing. Camera-based bill assessment requires a measured evaluation before any accuracy or counterfeit-detection claim. Software implementation and test procedures alone do not establish field readiness.

# Target Market

Our initial users are UM Tagum students and staff who need usable cash denominations or access between physical cash and an existing wallet. Early adopters would be people who report recurring difficulty with these tasks and accept a clearly disclosed fee. Their number and transaction frequency remain unmeasured.

| Participant           | Proposed relationship                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------|
| Students and staff    | Service users and transaction-fee customers.                                              |
| Campus administration | Proposed host and deployment approver. No host commitment is claimed.                     |
| Nearby merchants      | Secondary beneficiaries and interview participants. No merchant partnerships are claimed. |
| Coinnect team         | Proposed operator responsible for cash replenishment, maintenance, and customer support.  |

## Alternatives and differentiation

Users may ask a cashier or acquaintance for change, use an available cash-in or cash-out outlet, or rely on other payment options. Interviews and a local service review will establish actual availability, charges, and convenience. Coinnect's proposed distinction is a campus location combining denomination conversion and wallet access, with local cash conversion during internet interruptions. Convenience and cost advantages are hypotheses. We claim no exclusive technology or established competitive moat.

## Proposed campus pilot

We propose one supervised kiosk in an approved indoor, climate-controlled campus location. The team would agree operating hours, operator responsibilities, cash custody, and a support contact with the host. Initial testing would focus on cash conversion. Wallet transactions would follow only after the relevant readiness checks. The pilot is not an approved deployment or an existing commitment by UM.

| Stage                  | Evidence needed before advancing                                                                         |
|------------------------|----------------------------------------------------------------------------------------------------------|
| Discover demand        | Interview findings and a review of existing alternatives support a specific use case.                    |
| Verify the kiosk       | Recorded intake, payout, inventory, and interruption tests show explainable transaction outcomes.        |
| Run a supervised trial | Host permission, a reconciled cash float, visible fees, a support process, and a ready operator.         |
| Review continuation    | Observed use, fees accepted, service reliability, user feedback, and operating cost support a next step. |

## Pilot measures

Track attempted and completed transactions by service, repeat use with consent, completion time, inventory shortages, support requests, and unresolved claims. Compare users' reported time and total cost with their previous alternative. Record results without presenting pilot targets as achieved performance.

# Business Model and Incubation Request

## Transaction fee revenue

We propose a team-operated kiosk that earns disclosed service fees on completed transactions. Fees would be evaluated through user interviews, local service comparisons, and operating-cost measurement. We have not validated pricing, demand, or profitability. Cash deposited or dispensed is transaction principal and must not be counted as revenue.

For planning, monthly fee revenue equals the sum of completed transactions for each service multiplied by its fee. Contribution equals fee revenue less provider charges and other direct transaction costs. Operating surplus then deducts site expenses, maintenance, connectivity, power, support, and other recurring costs. A break-even estimate is meaningful only after these inputs and the mix of services have been measured.

## Costs and cash requirements

Startup needs include the enclosure, controllers, cash handling hardware, screen, camera, printer, assembly, and validation. Operating needs include cash collection and replenishment, spare parts, receipt paper, maintenance labor, support, and any host charges. The kiosk also needs a funded cash float and, where applicable, provider settlement funds. These liquidity needs are separate from revenue and hardware costs. Repository component estimates are not a verified cost for a finished kiosk.

## Social benefit to test

We aim to make appropriate cash denominations and wallet access more convenient for campus users and reduce avoidable trips or requests for change. Nearby merchants may benefit when customers arrive with suitable denominations. The pilot should measure whether these benefits occur, whether fees remain acceptable, and whether users can complete tasks with limited assistance. No inclusion impact or savings figure has yet been established.

## Incubation support requested

We seek mentorship in customer discovery, fee testing, and kiosk operations; technical feedback on cash handling and failure recovery; and introductions for a potential campus pilot. We also seek guidance on the provider and compliance requirements that apply before live financial services. Access to relevant testing resources would help us turn a partial prototype into evidence suitable for a deployment decision.

## Expected pre incubation outputs

Our proposed outputs are documented user needs, a validated first service scope, a physical test record, an operating-cost model, and a pilot proposal with clear operator responsibilities. We would use those results to decide whether to refine the prototype, run a supervised campus trial, or change the business assumptions. Any live launch remains contingent on readiness and appropriate approvals.

Evidence basis: team-supplied identity and prototype status; Coinnect repository architecture, implementation, and UAT procedures; organizer posters for submission requirements. This application makes no external market-size or regulatory-status claims.